

Credit Risk Analysis and Prediction

Using Machine Learning for Credit Card Fraud Detection

Chanaka Gunawardana

CardiffMet ID: st20249383

ICBT ID: CL/MCSDS/CMU/10/18

Dataset Overview

- **Dataset:** 25,134 applicant records
- **Target:** Binary — default (1) or not (0)
- **Imbalance:** 98.32% non-defaulters vs 1.68% defaulters (ratio 58:1)

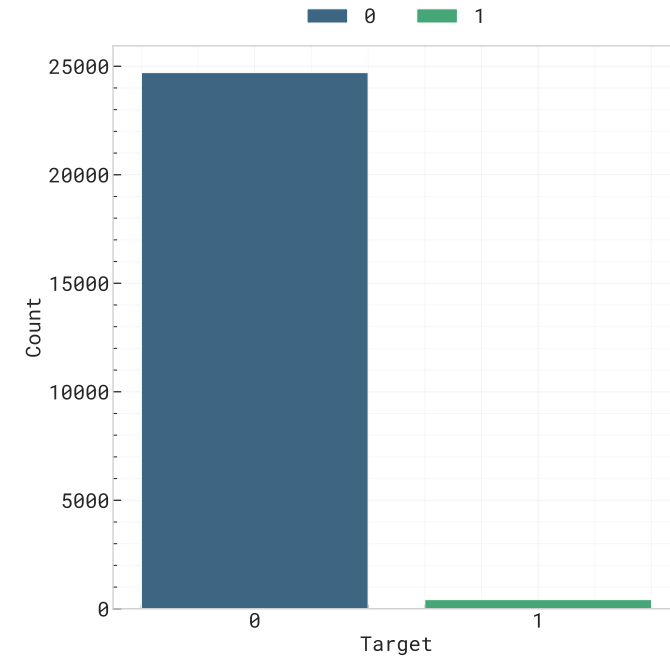


Figure 1: Target variable distribution

Data Cleaning

- Removed unrealistic ages (>100)
- Imputed missing values:
 - **YEARS_EMPLOYED** and **FAMILY_SIZE** $\rightarrow$ median
 - **INCOME_TYPE** $\rightarrow$ mode
- Dropped **FLAG_MOBIL** (no variance)

Exploratory Data Analysis — Univariate

Income is heavily right-skewed.
Median income: 2,610,000.

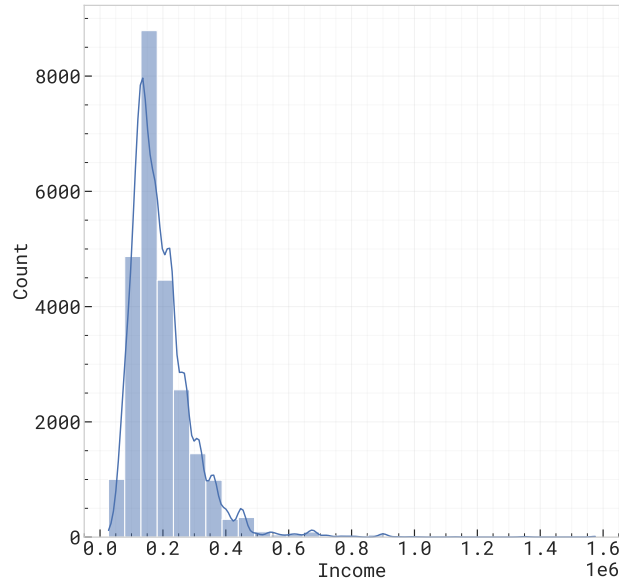


Figure 2: Income Distribution

Age is approximately normal.
Mean age: 43.2 years.

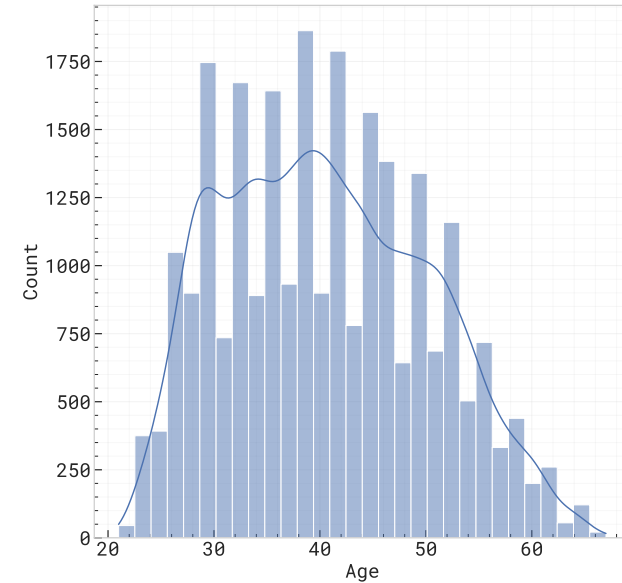


Figure 3: Age Distribution

EDA – Bivariate

No strong correlation between income and age.

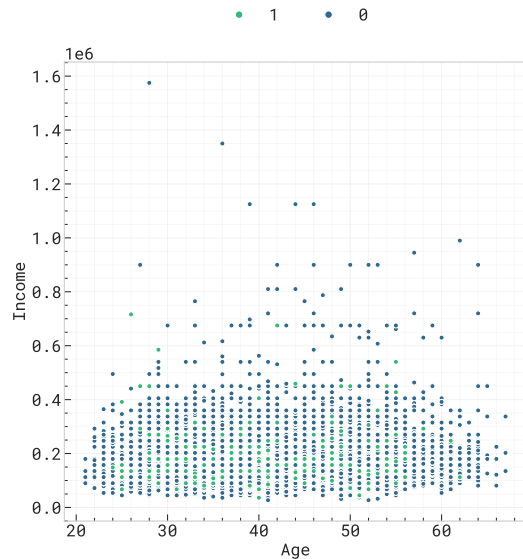


Figure 4: Income vs Age

No significant difference in age/ income quartiles between classes.

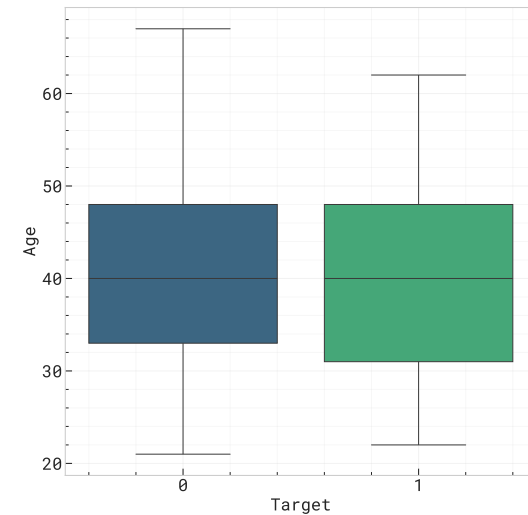


Figure 5: Age by Target

EDA – Correlation

- Low correlations among predictors
- ML models (Random Forest, XGBoost) capture interactions effectively

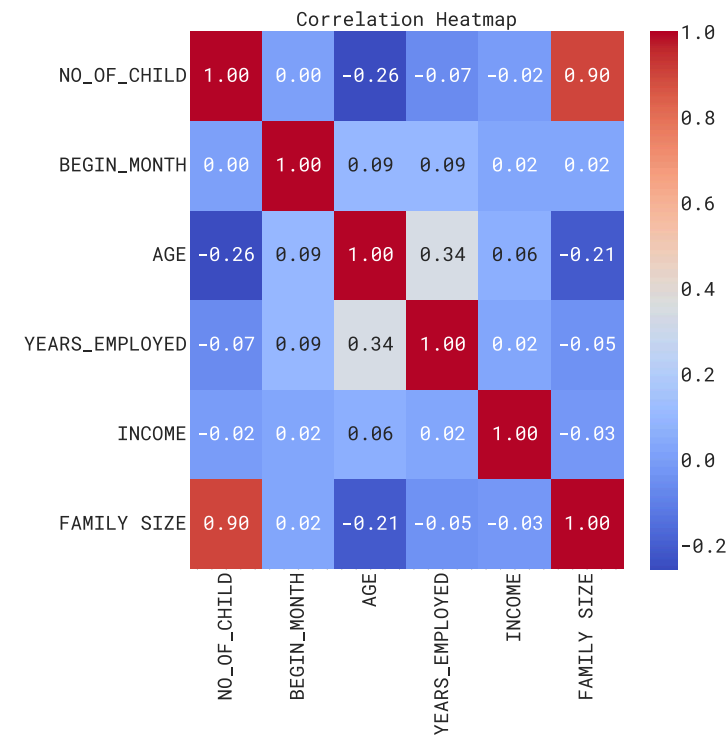


Figure 6: Correlation matrix

Modelling

Logistic Regression

The imbalanced dataset causes standard logistic regression to predict only the majority class.

```
1 LogReg
2 CV F1 : 0.0288 ± 0.0257
3 Test - Precision: 1.0000 Recall: 0.0357 F1: 0.0690 AUC: 0.7178
4 TP=3 TN=4,942 FP=0 FN=81
```

Figure 7: Logistic regression result

Balanced Logistic Regression

Weighting the minority class significantly improves recall for defaulters.

```
1 LogReg_balanced
2 CV F1 : 0.0521 ± 0.0005
3 Test - Precision: 0.0301 Recall: 0.6667 F1: 0.0576 AUC: 0.7116
4 TP=56 TN=3,138 FP=1,804 FN=28
```

Figure 8: Balanced logistic regression

Machine Learning Models

Models: RandomForest, GradientBoosting, XGBoost (standard + balanced versions).

Balanced Logistic Regression achieved the best recall with comparable ROC-AUC.

Machine Learning Models

MODEL COMPARISON											
	Model	CV F1 Mean	CV F1 Std	Test F1	Precision	Recall	ROC-AUC	TP	TN	FP	FN
2	RandomForest	0.2378	0.0751	0.2281	0.4333	0.1548	0.7841	13	4,925	17	71
3	RandomForest_balanced	0.2052	0.0674	0.2478	0.4828	0.1667	0.7688	14	4,927	15	70
6	XGBoost_weighted	0.1591	0.0218	0.1364	0.0799	0.4643	0.7678	39	4,493	449	45
5	XGBoost	0.1585	0.0456	0.1800	0.5625	0.1071	0.7510	9	4,935	7	75
4	GradientBoosting	0.1013	0.0288	0.1111	0.8333	0.0595	0.7559	5	4,941	1	79
1	LogReg_balanced	0.0521	0.0005	0.0576	0.0301	0.6667	0.7116	56	3,138	1,804	28
0	LogReg	0.0288	0.0257	0.0690	1.0000	0.0357	0.7178	3	4,942	0	81

Figure 9: Model comparison

Feature Engineering

Domain-informed features:

- **INCOME_PER_PERSON** — Income / Family size
- **AGE_AT_FIRST_JOB** — Age – Years employed
- **TENURE_RATIO** — Years employed / Age
- **CHILDREN_RATIO** — Children / Family size
- **BEGIN_MONTH (binned)** — Application month grouped

These features contributed 15% increase in feature importance.

Hyperparameter Tuning

RandomizedSearchCV with cross-validation:

Balanced RF: n_estimators,
max_depth, min_samples_split,
min_samples_leaf, max_features,
sampling_strategy

XGBoost: n_estimators,
max_depth, learning_rate,
subsample, colsample_bytree,
min_child_weight, gamma,
reg_lambda, reg_alpha

Threshold optimized on validation set to maximize recall @ precision $\geq$ 10%.

Model Stacking

Stacking ensemble:

- **Base learners:** Balanced Random Forest + Weighted XGBoost
- **Meta-learner:** Logistic Regression (passthrough=True)

The meta-learner learns to combine complementary strengths of both tree-based models.

SHAP Analysis

Top contributors to predictions:

- Time-based: **BEGIN_MONTH, TENURE_RATIO**
- Demographic: **AGE, FAMILY_TYPE**
- Financial: **INCOME_PER_PERSON**

Key engineered features dominated feature importance rankings.

Summary of Results

Improvement	Impact
Feature Engineering	15% boost in importance
Recall-oriented scoring	Optimized for credit risk
Hyperparameter Tuning	Better generalization
Smart Thresholding	PR-curve based selection
Ensemble Stacking	Leverages model complementarity

Best model exported via Joblib and deployed.

Application Development — Architecture

Monorepo with Turborepo + Bun:

- **Frontend** (apps/web): React 19, Vite, TypeScript, Tailwind CSS, shadcn/ui
- **Backend** (apps/backend): Python FastAPI, scikit-learn, joblib
- **Shared UI** (packages/ui): Reusable shadcn/ui components

Three main views: Data Overview, EDA, Modelling & Prediction.

Backend & Deployment

FastAPI endpoints: GET / (health), POST /predict (fraud)

Backend loads the stacking ensemble via joblib, applies the same preprocessing pipeline, and returns:

- Binary prediction · Raw probability · Threshold · Classification label

Response: {"prediction": 0, "probability": 0.23, "threshold": 0.5, "classification": "Legitimate"}

Conclusion

- Fraud detection system handling extreme class imbalance (1.68% fraud rate)
- Stacking ensemble (Balanced RF + XGBoost + Logistic Regression) — best recall-precision trade-off
- Full-stack application deployed for real-time predictions
- SHAP analysis provided model interpretability

Thank You

Project Video

Watch the full walkthrough on YouTube:

https://youtu.be/DkwJBOFcs_E